UNIVERSITI TEKNOLOGI MARA FACULTY OF INFORMATION SCIENCE

CAMPUS PUNCAK PERDANA, SELANGOR

BARCHELOR OF INFORMATION SYSTEM MANAGEMENT

(CDIM262)

GROUPING ASSIGNMENT (50%):

WEB APPLICATION

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Acknowledgement

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1.0 INTRODUCTION

The Car Rental system is a web-based application that developed to manage the process of rental's vehicle operations. The system digitizes manual processes through its structured platform which enables users to access available vehicles while handling their bookings and payment processes and tracking their rental activities. The system provides administrators with secure methods to control user accounts and rental operations and vehicle inventory and payment information.

The primary purpose of this system exists to enhance the car rental experience through automated processes which decrease human mistakes and boost operational efficiency through database integration. The system uses relational database management systems to store and retrieve essential information which includes vehicle status rental information customer data and payment information. The system protects data integrity through its authentication and authorization system which prevents unauthorized access and enables users to access only their authorized information.

The Car Rental System development goals of IMS566: Advanced Web Design Development and Content Management course support its academic objectives. The document demonstrates the process of using database connections together with content management methods to develop user interfaces and user experiences which support CRUD operations. The project develops technical system development skills while teaching students to work as a team through documentation and their ability to develop functional web solutions from abstract ideas.

The system provides an integrated platform which ensures precise data handling while streamlining customer and administrator operations through its application of fundamental web design and database management principles.

2.0 GITHUB REPOSITORY LINK

The Car Rental System web application source code exists on GitHub for developers to work together and manage different versions of the software. The repository contains all project source files together with database scripts and all related documentation.

Github Repository Link:

https://github.com/adamltf44/CarRental.git

3.0 ENTITY-RELATIONSHIP DIAGRAM (ERD) - isha

The Entity-Relationship Diagram (ERD) shows both the database structure of the Car Rental System and its data entity relationships. The car_rental_db relational database of the system contains five main entities which include Users, Roles, Cars, Rentals, and Payments. The organisations exist to support payment monitoring, car management, rental processing, and authentication activities. The ERD uses primary keys and foreign keys to achieve data normalisation while reducing data duplication and maintaining data integrity.

3.1 Entity Description

1. Users

User's table stores information that related to the system like customers and administrators. It is used for authentication and authorization purposes.

Attributes:

- user_id (PK)
- role_id (FK)
- full_name
- email
- password
- phone_number
- created_at
- modified
- Status

2. Roles

Roles is to define access levels within the system like administrator and customer.

Attributes:

- role_id (PK)
- role_name
- status
- created_at
- Modified

3.Cars

Car’s table table stores detailed information about the cars that availale for rental

Attributes:

- car_id (PK)
- category_id
- brand
- model
- year
- fuel_type
- drive_type
- transmission
- engine_cylinders
- displacement
- plate_number
- daily_rate
- status
- category_name
- description
- created_at
- modified

4. Rentals

Rental’s table records car booking details made by users

Attributes:

- rental_id (PK)
- user_id (FK)
- car_id (FK)
- start_date
- end_date
- total_days
- total_price
- rental_status
- created_at
- modified
- Status

5. Payments

Payment's table stores payment details for car that is out for rent.

Attributes:

- payment_id (PK)
- rental_id (FK)
- user_id (FK)
- payment_methods
- payment_date
- amount
- payment_status
- created_at
- Modified

3.2 Relationship Between Entities.

- **Roles (1) — (M) Users**
  One role can be assigned to many users, but each user has only one role.
- **Users (1) — (M) Rentals**
  A user can make multiple rental bookings, but each rental belongs to only one user.
- **Cars (1) — (M) Rentals**
  A car can appear in many rental records over time.
- **Rentals (1) — (M) Payments**
  Each rental may have one or more payment records.
- **Users (1) — (M) Payments**
  A user can make multiple payments.

4.0 SYSTEM REQUIREMENTS

The CarRental web application requires spesific hardware and software components to run correctly. These requirements ensure the system runs smoothly during the development, testing, and deployment.

Software Requirements

- **Operating Systems:** Windows10 / 11, macOs, or Linux
- **Web Server:** Apache (via XAMPP, WAMP, MAMP, LAMP)
- **Programming Language:** PHP8
- **Database:** MySQL
- **Development Tools:** Visual Studio Code, Git
- **Utilities:** phpMyAdmin for database administration

Version Compatibility

To provide consistent speed and security protection, the system has been designed and tested to work with PHP, Apache, MySQL, and modern web browsers.

Dependencies

Database Dependencies:

- **MySQL Server –** For storing user, vehicle, booking, and transaction data
- **phpMyadmin –** For database management and administration

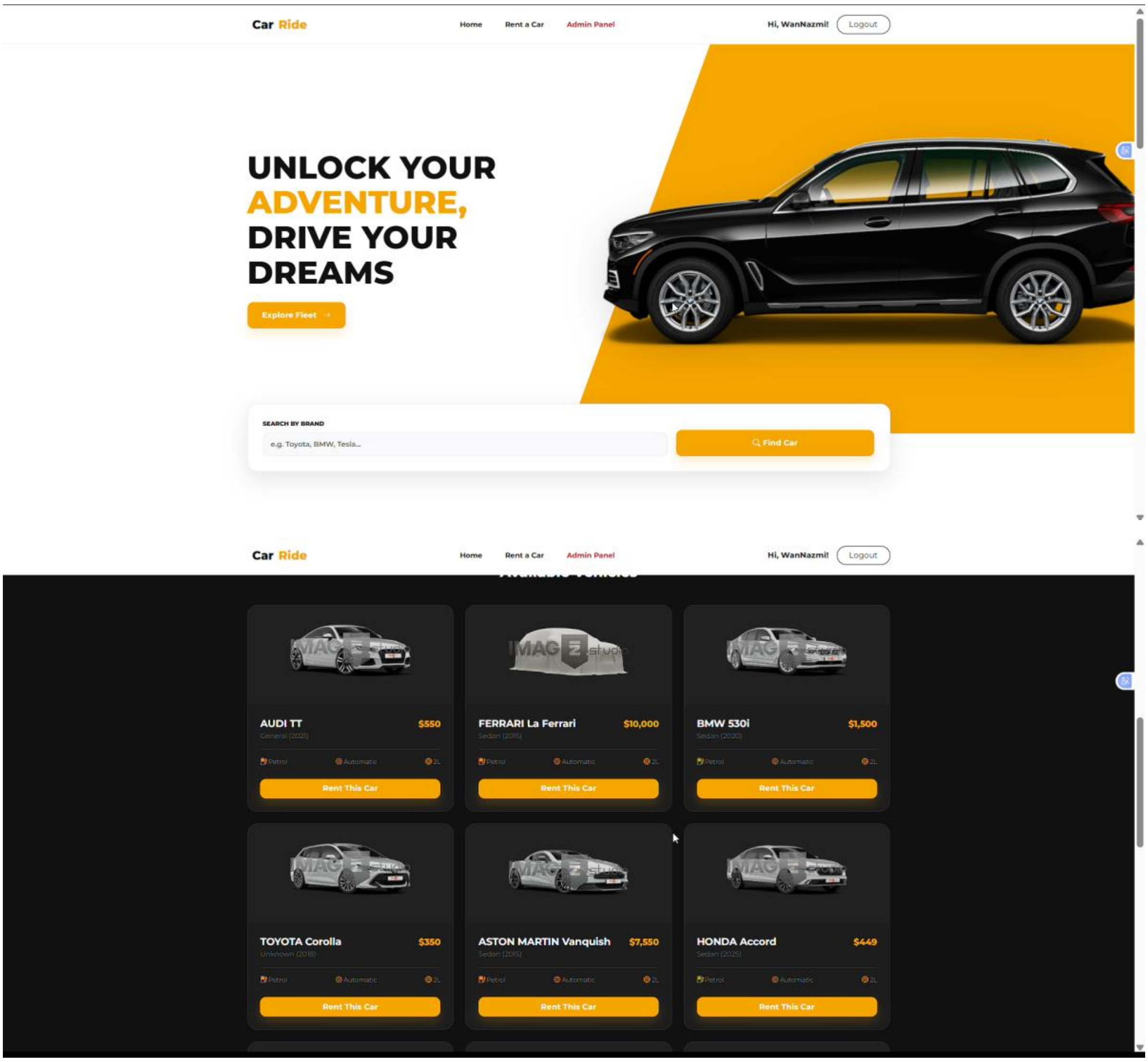
Frontend Dependencies:

- **Bootstrap 5 –** For responsive user interface design
- **HTML & CSS –** For structure and styling
- **Javascript –** For form validation and interactive elements

5.0 USER INTERFACE OVERVIEW

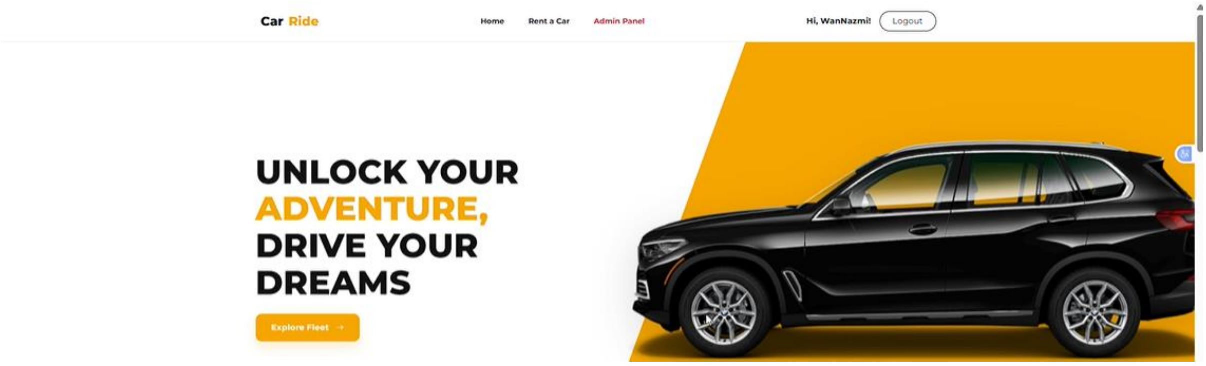
This section will focus on the overall layout, visual design, and navigation structure of the Car Rental System website. The user interface will be clean, intuitive, and user-friendly to ensure a smooth user experience for both customers and administrators.

Homepage Layou



The homepage is the main entry point of the system. It provides a good overview of the car rental service and allows users to access various key functions of the system quickly. The homepage has three main sections: a navigation bar at the top, a main content area in the center, and a footer at the bottom of the page. The main content area of the homepage displays information regarding the cars available, offers, and call-to-action buttons such as “Search by Brand” and “Rent Car”.

Navigation Bar

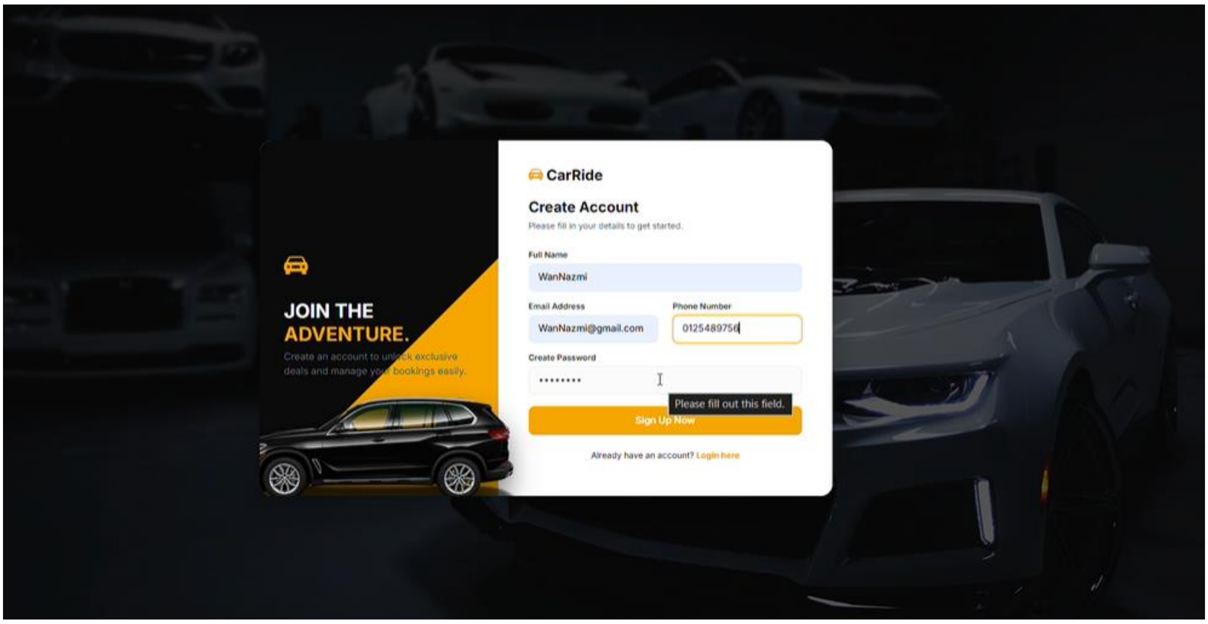


The navigation bar is located at the top of the website and is consistent across all pages. The navigation bar gives access to the different sections of the system, such as Home, Rent a Car, Booking, Admin Panel, and Logout. The consistency of the navigation bar ensures that the user can navigate through different pages with ease without any confusion.

6.0 FEATURES & FUNCTIONALITIES

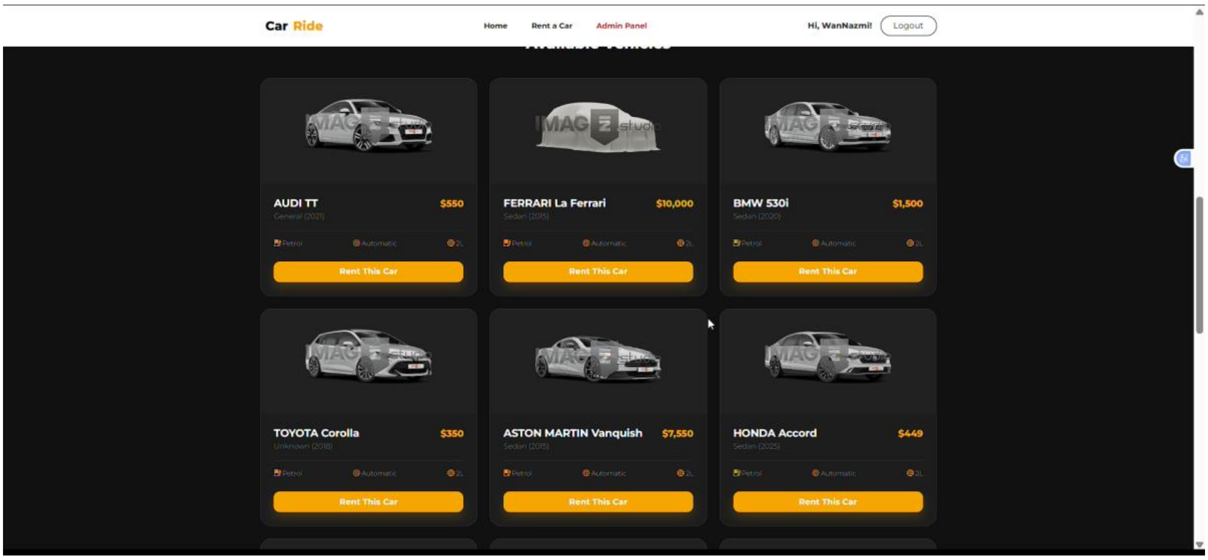
This section describes the main features and functionalities associated with the Car Rental System. Each feature has been incorporated to facilitate the main business needs of running a car rental business while at the same time offering convenience to users.

User Registration and Login



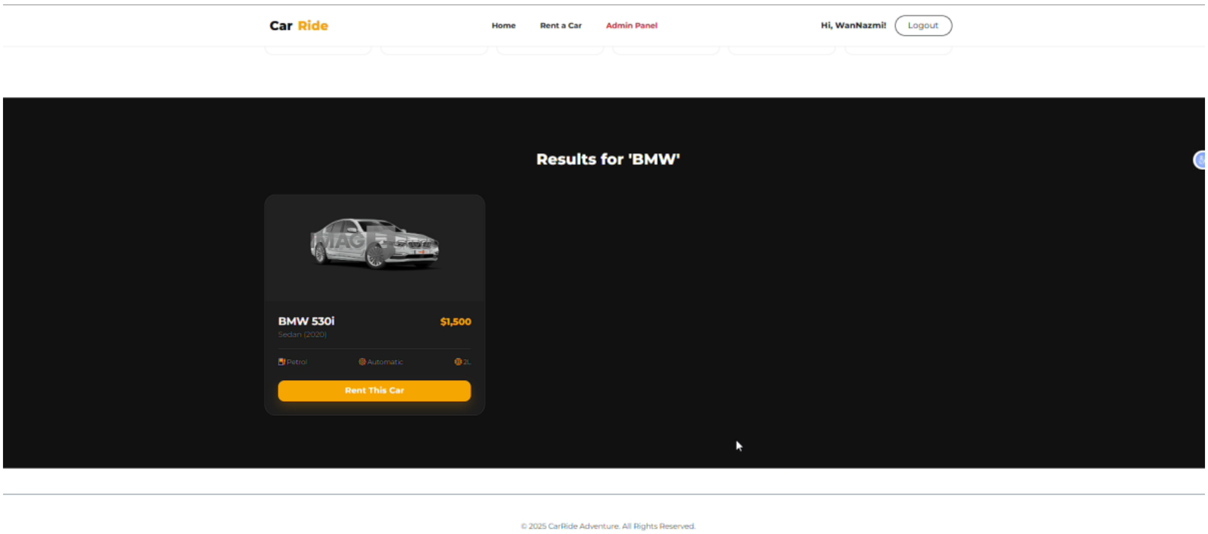
The system offers customer authentication features that allow users to create accounts and authenticate themselves before making bookings. A new customer can create an account by providing basic information about themselves, while an existing customer can authenticate themselves by entering their details. This feature ensures that only authenticated users are allowed to make bookings.

Car Listing and Availability Display



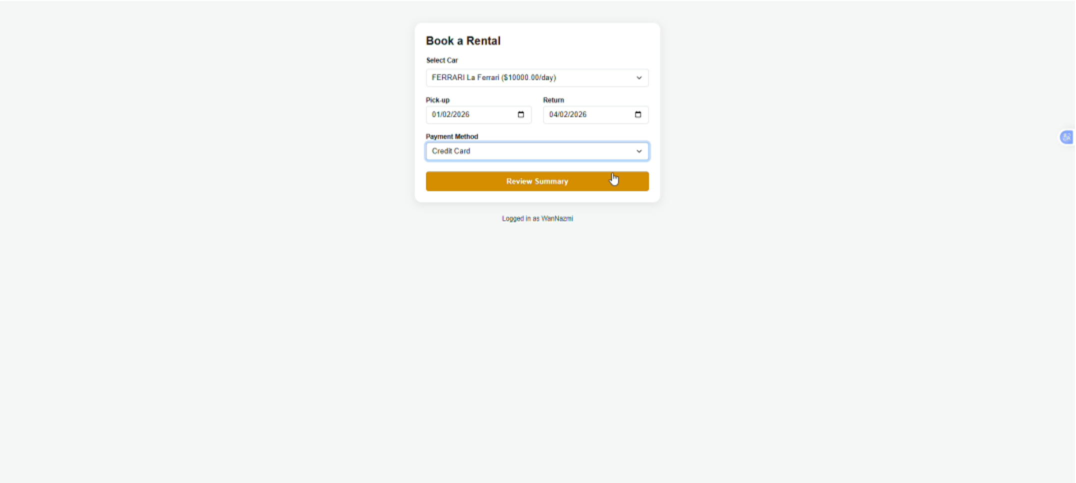
The car listing feature provides a comprehensive display of all available cars in the system. The display includes key information regarding the car, such as its model, type, rental price per day, and availability status.

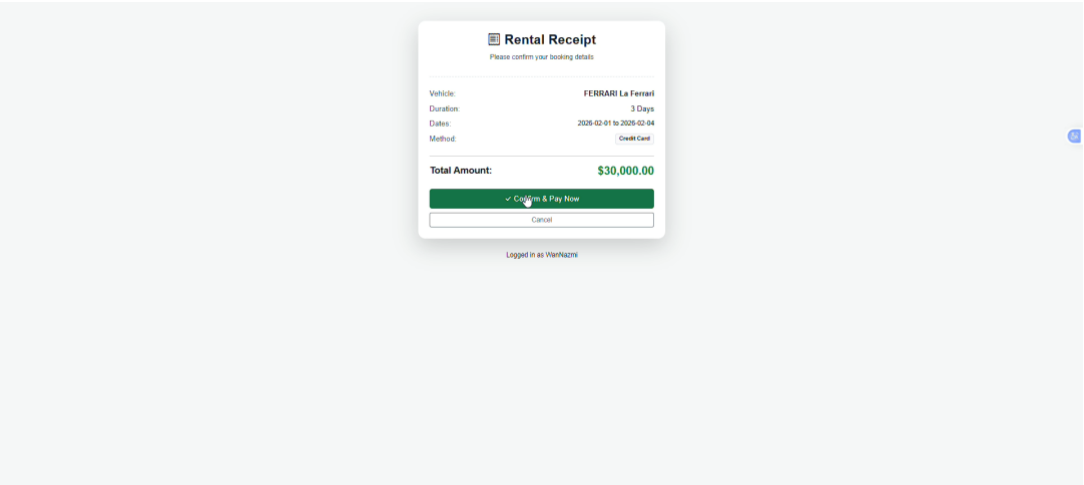
Car Details View



Users have the ability to view detailed information for the chosen car. This page offers more information, including specifications, conditions for renting, and pricing for the car. This will enable users to make informed decisions according to their preferences and requirements.

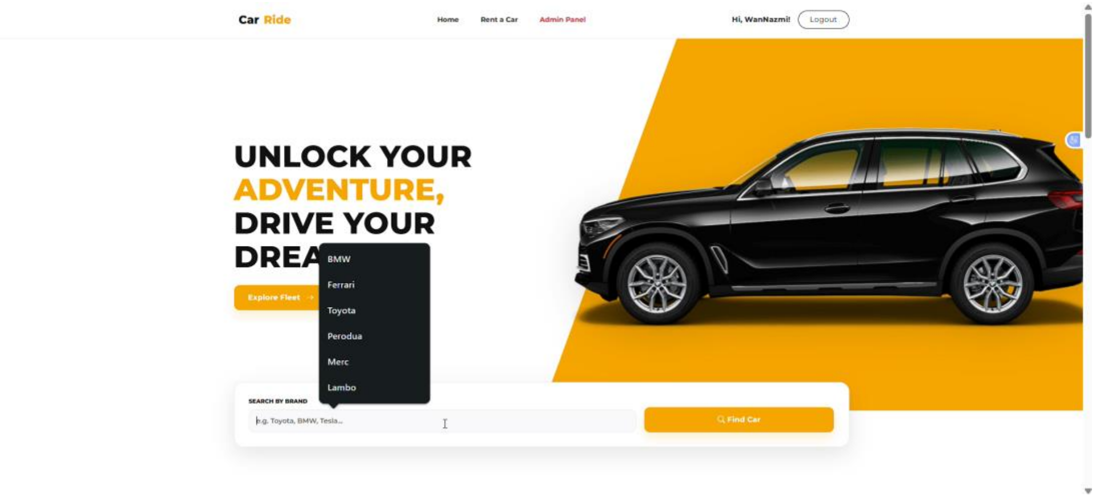
Booking Management





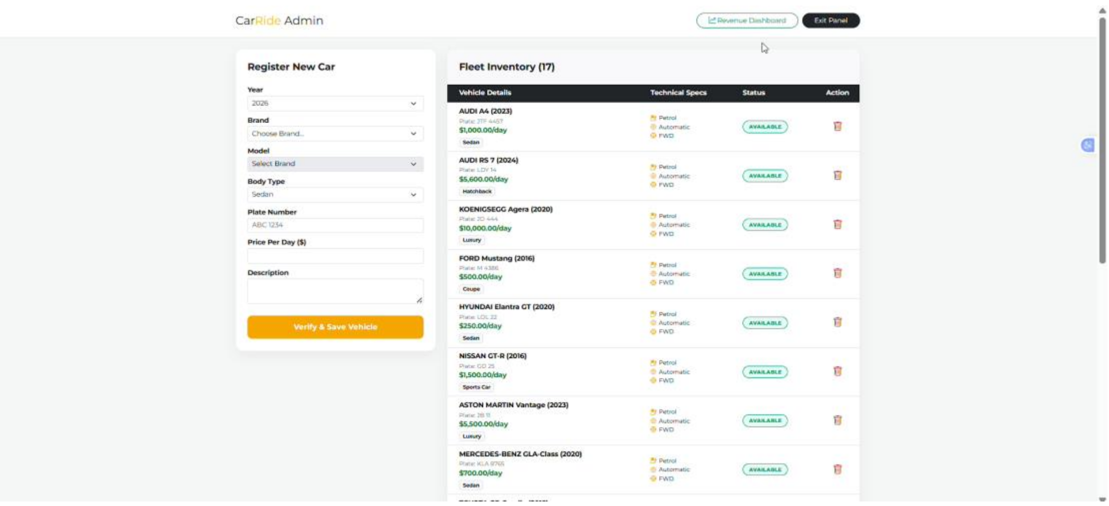
The feature for booking enables users to book a car, set the dates for the car rental, and send a request for booking. The system will then store the details of the booking and will generate a confirmation receipt in the database, updating the status of the car accordingly.

Search and Filter Functionality



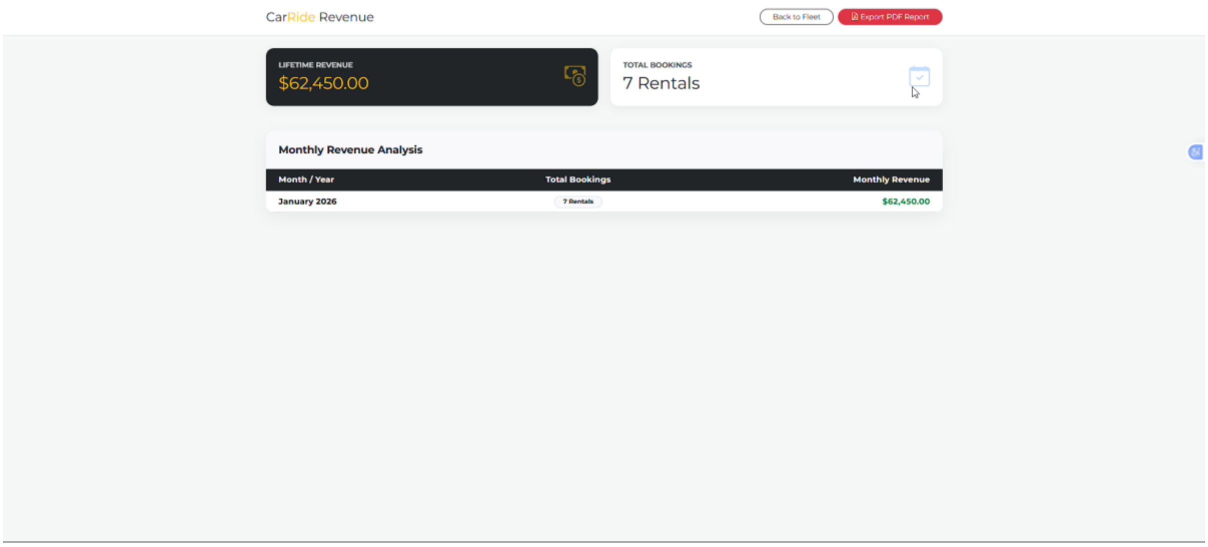
The system has search and filter options that enable the user to search for cars according to certain criteria, such as car brand. This feature enhances usability by shortening the time required to search for appropriate cars.

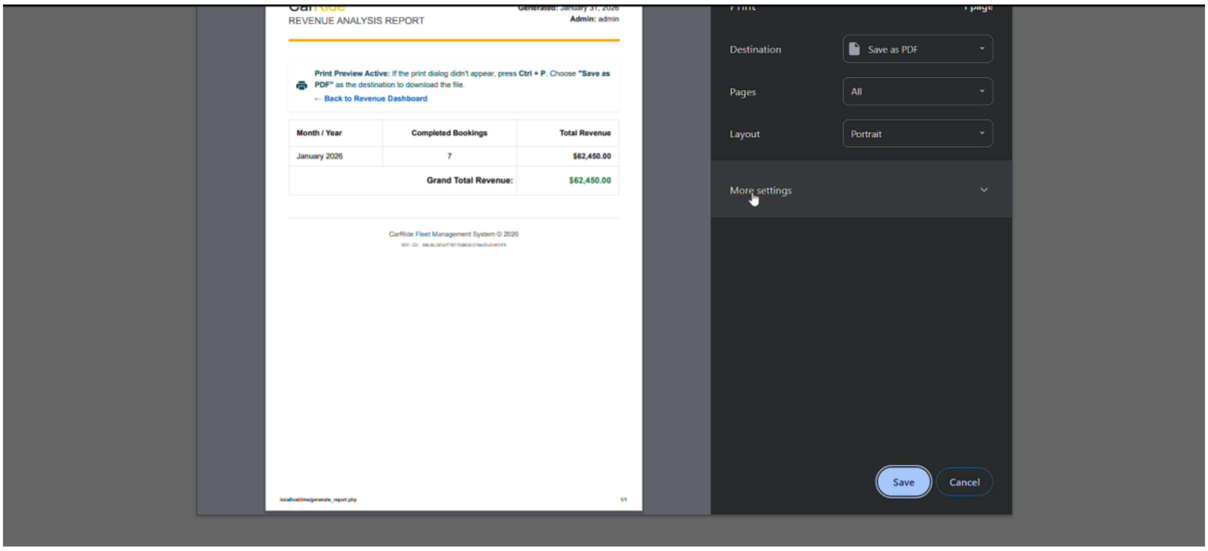
Admin Dashboard



The admin dashboard is used by the administrators to offer management functions. The admin users are able to create new cars, edit existing car details, remove unavailability of cars, and view all bookings made by users.

Report Generation (PDF Export)





The system supports report generation by allowing administrators to export booking records into PDF format. This feature is useful for documentation, reporting, and record-keeping purposes.

In conclusion, the Car Rental System integrates essential features and functionalities to support both user convenience and administrative efficiency, ensuring a reliable and effective car rental management process.

7.0 WORKFLOW OF FORM

The user begins the form process after opening the CarRental web application which displays the login interface to user. The user needs to provide valid credentials for system entry. The user reaches the main dashboard after finishing the authentication.

The user can access the car booking form from the dashboard. The booking form requires the user to enter their pickup date and return date and car type information along with their personal data. The user submits the form after completing all required fields. The system checks input validation to verify whether users entered all mandatory information correctly. The system stores booking information in the database when validation succeeds **(Create).**

The user receives a confirmation page which displays all their bookings after they successfully submit their information. The system allows users to access all booking details which include options to see booking information **(Read)** and to make necessary changes **(Update)** and to cancel their reservations **(Delete)**. The user may also search for specific bookings by criteria such as date or car type to quickly locate records.

The system allows users to export their booking information as PDF files which they can use for record-keeping and reporting purposes. The export feature enables users to create a downloadable summary document which contains information about their bookings. The system protects data integrity throughout the workflow process while granting form access permissions only to verified users.

8.0 TEAM ROLES & CONTRIBUTIONS

Name	Role	Contribution
ADAM BIN MOHD LATIFF	Project Manager	Coordinated group tasks, managed timeline, ensured project requirements were met, and led final integration
WAN NAZMI MUQRI BIN WAN AHMAD NASIM	Backend Developer	Developed CRUD operations, database connection, authentication, and server-side logic
AMMAR RAFIQ BIN ABDUL RAZAK	Frontend Developer	Designed user interface using Bootstrap/Tailwind, ensured responsive layout and navigation
NOOR ANISHA BINTI HASAN	Database Designer	Designed ERD, created database schema, and handled relationships between tables
NUR UMAIRAH BATRISYIA BINTI NOORZAM	Documentation & Testing	Prepared system documentation and user manual, tested system functionality and fixed minor bugs

9.0 CONTACT INFORMATION (SUPPORT)

For any system issues, technical support, or inquiries regarding the application, users may contact:

- **Email:** ammarrafiqas@gmail.com
- **Project Manager:** Adam Latiff
- **GitHub Repository:** https://github.com/adamltf44/CarRental.git
- **Id:** admin
- **Password:** 123456789

Support is available during normal working hours (9.00 AM – 5.00 PM).

10.0 CONCLUSION & REFLECTION

In conclusion, the development of the Car Rental System was a success in the sense that it was able to achieve its main goal of transforming a manual vehicle rental system into a fully functional web-based application. This was achieved through the implementation of a number of fundamental features within the system, all supported by a structured relational database design. This not only improved the efficiency and effectiveness of the system but also reduced human error and provided a better overall user experience for all users.

In terms of reflection, this project provided a valuable hands-on experience in applying theoretical knowledge learned from the IMS566 module to a real-world project. From a practical perspective, the project provided a deeper understanding of a number of fundamental aspects within a web-based application, including the use and implementation of a relational database design, the planning and implementation of a system workflow, and the implementation of a number of fundamental features within a web-based application, including CRUD operations and secure authentication and authorization mechanisms.

In addition, this project also provided a valuable learning experience in the importance of teamwork and effective time management skills within a project environment. In terms of teamwork, the project provided a number of challenges, including the integration of a number of system components, debugging issues within the project, and balancing individual work contributions to meet project requirements. However, through effective teamwork and problem-solving skills, the project was able to achieve its goals and objectives within the expected timeframe and provided a fully functional web-based application.

In conclusion, this project was a valuable learning experience in a number of aspects, including problem-solving skills, teamwork, and effective time management skills. In terms of future use, the knowledge and experience gained from this project have provided the team with a solid foundation in web-based application development and have prepared us to handle a number of future system development tasks and projects.